RESEARCH ARTICLE

When Does Posterior Integration Change Option Values? Evidence from WTI Average Price Options

Anonymous Author[1]

Abstract

We examine how volatility uncertainty and volatility estimation affect WTI Average Price Option (APO) values. Posterior-integrated pricing (PI) averages option values over the volatility posterior; posterior-mean plug-in pricing (PM) uses its mean. In 10 million simulated return histories, PI–PM differences depend on posterior variance and price curvature. For 310 October 2026 contract-date observations, the mean absolute PI–PM difference is $0.000745 per barrel, despite a mean 95% posterior model-price interval width of $0.2733. Using 2,381 observations across 15 valuation dates, earlier option-implied volatility gives lower error against settlement-price proxies than historical volatility (mean absolute error $0.1075 versus $2.6187 per barrel). This persists when carrying forward each contract's last implied volatility without fitting a smile. These conditional comparisons use valuation-date futures inputs and a one-factor model. Sparse trading, unverified vendor timestamps, and futures-correlation assumptions limit interpretation; no trading advantage is established.

KEYWORDS

Bayesian option pricing; Asian options; WTI crude oil; parameter uncertainty; implied volatility

1 | INTRODUCTION

Bayesian option valuation can account for parameter uncertainty by integrating conditional prices over a posterior distribution rather than using a point estimate (Guidolin and Timmermann 2003, Rombouts and Stentoft 2014, Gupta and Reisinger 2014). The effect depends on price curvature and the spread of the volatility posterior. We study this question for WTI Average Price Options (APOs), focusing on uncertainty about volatility estimated from historical returns.

Arithmetic-average Asian option values are sensitive to moneyness, maturity, and the proportion of the average already fixed. Established methods exploit geometric-average control variates or conditioning approximations (Kemna and Vorst 1990, Curran 1994). Commodity applications must also account for the futures term structure; richer models can incorporate stochastic convenience yields, jumps, and stochastic volatility (Schwartz 1997, Shiraya and Takahashi 2011, Ewald et al. 2023). We hold the one-factor pricing model fixed and compare two effects: averaging prices over the volatility posterior and choosing between historical and option-implied volatility.

The two comparisons answer different questions. The posterior-integrated price (PI) averages conditional option values over the volatility posterior, whereas the posterior-mean plug-in price (PM) uses posterior mean volatility. Their difference measures how much posterior integration changes the point price. It does not measure the full uncertainty in model prices. The second comparison holds the pricing model fixed and asks which volatility estimates give smaller errors against settlement-price proxies. Because we also use the futures curve and known fixings from the valuation date, this is a conditional pricing comparison, not a fully prospective forecast.

We study the PI–PM difference using 10 million simulated return histories and WTI APO observations from October 2026. We compare historical volatility with implied volatility from earlier APO observations across 15 valuation dates. We also repeat the comparison using implied volatility from standard monthly WTI futures

options rather than APOs. Implied-volatility estimates use only earlier option observations; repricing uses the valuation-date futures curve, realized fixings, and discount factor.

The synthetic results show that posterior integration can matter when historical samples are short and the pricing function has substantial curvature in volatility. In the October 2026 sample, PI and PM are nearly identical, although uncertainty in the resulting model prices is much larger than their difference. Implied volatility from earlier option prices also gives lower pricing errors than the historical benchmarks tested, including rolling windows, EWMA, and horizon-matched GARCH(1,1). The result holds both without fitting an implied-volatility smile and when implied volatility comes from standard WTI futures options.

Unlike studies that compare stochastic-volatility or jump models, such as Cummins and Esposito (2025), we keep the pricing model unchanged. This lets us study volatility estimation, posterior averaging, and the resulting uncertainty in arithmetic-average option prices separately. The comparisons use the same pricing model throughout. They do not establish a trading advantage or identify a structural difference between physical and risk-neutral volatility. The methods and sample restrictions are described below; detailed numerical and robustness results appear in the online supplement.

2 | POSTERIOR UNCERTAINTY AND ASIAN-OPTION PRICING

2.1 | Physical inference and risk-neutral valuation

Historical log returns are modeled under the physical measure as

$$R_i \sim \mathcal{N}\left[\left(\mu - \frac{1}{2}\sigma^2\right)\Delta t, \sigma^2 \Delta t\right], \tag{1}$$

with baseline priors

$$\mu \sim \mathcal{N}(0,1), \qquad \sigma \sim \text{InvGamma}(2, 0.1), \qquad \sigma > 0. \tag{2}$$

We sample the posterior of $(\mu, \log\sigma)$ using random-walk Metropolis. For the large synthetic experiment, we integrate out μ analytically conditional on σ. We can then compute the one-dimensional posterior directly, without MCMC sampling error.

Pricing is carried out separately under the risk-neutral measure. For futures maturity u, the maintained benchmark is

$$dF_t^u = \sigma F_t^u \, dW_t^Q, \tag{3}$$

where u indexes futures maturity and the same Brownian motion W^Q drives all maturities. Under this model, futures of different maturities have perfect instantaneous correlation. Physical drift does not enter risk-neutral valuation. We keep these dynamics unchanged when comparing posterior integration and volatility estimates.

For an arithmetic-average option with fixing dates $t_1, \ldots, t_M$,

$$A_T = \frac{1}{M}\sum_{j=1}^{M} F_{t_j}^{(1)}, \tag{4}$$

and the conditional call value is

$$C_t^Q(\sigma) = B(t,T)\,\mathbb{E}^Q\left[(A_T - K)^+ \mid \mathcal{F}_t; \sigma\right], \tag{5}$$

where $B(t,T)$ is the discount factor, K the strike, and $\mathcal{F}_t$ the valuation-date information set. Put values are defined analogously. In Equation (4), $F_{t_j}^{(1)}$ is the first-nearby WTI futures settlement on fixing date t_j. At a valuation date within the averaging period, earlier fixings are known. We simulate only the remaining fixings, using the corresponding contracts in the observed CL futures curve as initial values.

We use Monte Carlo to price the baseline model with historical volatility. For the date-ordered conditional comparisons and checks using alternative constant volatilities, we use Curran's deterministic conditioning approximation (Curran 1994). We assess numerical accuracy using high-precision Monte Carlo and scrambled Sobol simulations.

2.2 | Posterior integration and price uncertainty

Let $\mathcal{D}$ denote the historical return data, $C^Q(\sigma)$ the conditional risk-neutral option value, and $\bar{\sigma} = \mathbb{E}[\sigma \mid \mathcal{D}]$. We compare posterior-integrated (PI) and posterior-mean plug-in (PM) prices:

$$C_{PI} = \mathbb{E}[C^Q(\sigma) \mid \mathcal{D}], \qquad C_{PM} = C^Q(\bar{\sigma}). \tag{6}$$

A second-order expansion around $\bar{\sigma}$ gives

$$C_{PI} - C_{PM} \approx \frac{1}{2} C^{Q\prime\prime}(\bar{\sigma}) \,\text{Var}(\sigma \mid \mathcal{D}). \tag{7}$$

This relates the PI–PM difference to posterior variance and price curvature. Its sign depends on local curvature, and it is most accurate when higher-order terms are small over the posterior support.

Uncertainty about volatility also leads to uncertainty about the model price:

$$C^Q(\sigma), \qquad \sigma \sim p(\sigma \mid \mathcal{D}). \tag{8}$$

Its dispersion is approximately

$$\mathrm{SD}\big(C^Q(\sigma) \mid \mathcal{D}\big) \approx \big|C^{Q\prime}(\bar{\sigma})\big| \sqrt{\mathrm{Var}(\sigma \mid \mathcal{D})}. \tag{9}$$

The PI–PM difference is approximately proportional to posterior volatility variance, whereas the standard deviation of model prices is approximately proportional to posterior volatility standard deviation. Neither quantity measures future market-price uncertainty or removes pricing-model misspecification.

2.3 | Synthetic study of posterior integration

The simulation varies posterior volatility uncertainty and price curvature, the two terms in Equation (7). It contains 10 million simulated return histories, generated with root seed 2468 and distributed over 35 combinations of historical sample size (21 to 1,260 observations) and annual volatility (0.10 to 0.80). We consider 168 option configurations with different maturities, levels of moneyness, and fractions of the average already fixed. For each history and option state, we calculate the PI–PM difference, posterior volatility variance, local price curvature, and the second-order approximation. We evaluate the volatility posterior on a 4,097-point grid and process the simulations in batches without changing the model or grid.

This allows us to assess the curvature–variance approximation without relying on observed market prices.

2.4 | Numerical reference and futures-correlation sensitivity

A separate synthetic check asks whether close PI and PM prices also imply small pricing errors against values computed at the true volatility. We generate sufficient statistics for Gaussian returns at two annual volatilities (0.20 and 0.50) and two historical sample sizes (21 and 252). For each combination, we generate 40 independent return histories and value calls with strikes of 95, 100, and 105 for a \$100 futures level. There are 160 independent return histories, with each history used for three strikes. We use a separate random seed (20261219) for this study. We chose the design after examining the pilot results, and it was not preregistered.

For each history, we integrate over the volatility posterior using a 513-point grid. This gives PI, PM, and a central 95% posterior interval for the model price under the one-factor approximation. We obtain an independent reference price at the true volatility using scrambled Sobol simulation and an exact geometric-average control variate. The reference reflects eight scheduled fixings drawn from four futures contracts. We consider three assumed correlations between futures contracts: $\rho \in \{1, 0.75, 0.50\}$. PI and PM are always priced using the one-factor model with $\rho = 1$. Only the independent reference price changes. Each simulated return history is used for three strikes and three assumed correlations. The resulting 1,440 option observations are not independent. We account for this dependence by resampling entire return histories within each volatility and sample-size group.

We measure the absolute error of the PM price against the independent reference price. We also report how often the posterior interval for the model price contains that reference. This measures coverage of a model value, not a future market price. We repeat the calculations with 513, 1,025, 2,049, and 4,097 volatility-grid points, keeping the simulated histories and independent reference prices unchanged. The number of reference prices covered remains unchanged, although some posterior interval endpoints shift by several thousandths of a dollar per barrel. This assesses numerical stability under the assumed model, not calibration to observed market prices.

2.5 | Research software verification and AI assistance

The calculations use version-controlled Python code and saved numerical results. The smaller synthetic study uses independent random draws and does not reuse the 10-million-history simulation. OpenAI ChatGPT assisted with Python implementation and debugging, literature organization, and manuscript drafting and editing. The author is responsible for verifying the code, statistical calculations, references, interpretations, and final manuscript.

3 | DATA AND EMPIRICAL DESIGN

3.1 | WTI Average Price Options

We use individual Barchart histories for WTI Average Price Options (Barchart 2026). The raw archive contains 213 unique contracts and 11,179 contract-date observations, with expiry months ranging from September 2026 to June 2029. We compare model prices with the Barchart end-of-day `Latest` field, used as a settlement-price proxy because official final settlement and trade-execution status have not been independently verified. We include available calls and puts and calculate moneyness separately for each valuation date.

For each option, we identify the known fixings and the futures contracts needed for the remaining fixings. We initialize those contracts with the futures prices observed on the valuation date. The first-nearby contract may change during the averaging month.

Historical volatility is estimated from WTI futures returns available by the valuation date. For historical return estimation, the baseline uses the continuous front-month `CL=F` series. A robustness check reconstructs a first-nearby return history contract by contract from CL settlements and excludes returns spanning contract switches.

3.2 | Date-ordered conditional pricing comparisons

We compare APO prices obtained using implied volatility recorded before each valuation date. We cannot independently verify when the vendor first published or later revised these records. For each price, we use the valuation-date futures curve, known fixings, and discount factor. Only the implied-volatility input comes from earlier option dates. Historical-volatility estimates follow the specified return-data cutoff. The comparison is therefore date-ordered and conditional, not a verified prospective forecast. The sample comprises 2,381 contract-date observations across seven expiries and 15 valuation dates.

The baseline averages conditional prices over the volatility posterior estimated from all available earlier returns. Historical alternatives use rolling windows of 63, 126, or 252 returns, an exponentially weighted moving average (EWMA) with a 63-business-day half-life, and a horizon-matched Gaussian GARCH(1,1) forecast. For GARCH, we estimate model parameters using earlier returns, update conditional variance using the valuation-date return, and forecast variance over the remaining fixing dates. Thus, this benchmark also uses some same-day information. We then find the constant annualized volatility that gives the same variance of the remaining average under the one-factor pricing model. Online Supplement Section S1 and Table S1 give the full results.

We obtain contract-level APO implied volatilities by inverting the Curran pricing approximation. For each expiry, the previous-day implied-volatility smile is estimated on the latest available earlier date. A second specification fits a smile using all earlier dates, giving more weight to recent observations. Both depend on moneyness and option type. Neither uses the target-date APO settlement to estimate volatility.

To test whether fitting an implied-volatility smile improves pricing, we also value each contract using its most recent implied volatility. We reprice with the valuation-date curve, discount factor, and fixing schedule. This same-contract benchmark covers 2,366 observations.

Pricing performance is summarized by mean error, mean absolute error (MAE), and root mean squared error (RMSE). We account for dependence among contracts on the same valuation date by resampling whole dates in a cluster bootstrap.

3.3 | Cross-instrument comparison

The first comparison estimates implied volatility from earlier APO observations. We therefore add a cross-instrument check using standard monthly American WTI futures options (CME product LO) from Databento (Databento 2026). We match LO option settlements to the underlying CL futures and invert prices using a Cox–Ross–Rubinstein futures-option tree.

For each APO target date, only LO observations from earlier dates are used. Implied-volatility smiles are fitted separately for the underlying CL futures contracts needed by the October-2026 APO fixing schedule. We convert these maturity-specific volatilities into one volatility input for the APO pricing model. The comparison covers 253 APO contract-date observations on 10 valuation dates. We include only contracts whose moneyness falls within the range observed in earlier LO data for both volatility specifications. Online Supplement Section S4 and Table S4 assess the volatility conversion and sensitivity to the number of steps in the Cox–Ross–Rubinstein tree.

This comparison asks whether the implied-volatility advantage also appears when volatility is estimated

from another WTI option product; it does not identify a variance risk premium.

4 | RESULTS

4.1 | When does posterior integration matter?

Figure 1 shows when posterior integration affects point prices in the simulated settings. The largest differences occur with short estimation windows, high volatility, long maturities, few realized fixings, and pronounced price curvature. Across the 10 million synthetic histories, the maximum observed absolute difference is 0.1565 for a normalized $100 forward, and the largest within-cell 99th percentile is 0.1273. These extremes exceed the differences observed in the October-2026 sample. The observed maximum depends strongly on simulation size and rare outcomes; the within-cell 99th percentile is less sensitive to them.

The second-order approximation tracks the simulated differences closely: the correlation between cell-level mean PI–PM differences and the curvature–variance term is 0.9995. Regressions through the origin also give slopes near one in the settings with the largest differences. The results indicate that posterior variance and price curvature largely explain the local PI–PM difference.

The PI–PM difference usually falls as more fixings become known. It does not always fall steadily. In Panel C, the 252-day curve initially rises because the additional fixings change volatility exposure and price curvature. The difference nevertheless tends toward zero as the averaging period ends.

4.2 | Pricing error and sensitivity to futures correlation

The separate simulation uses 160 independent return histories and 480 option observations for each assumed futures correlation. With perfect correlation ($\rho = 1$), the mean absolute PI–PM difference is $0.003709 per barrel. By comparison, the mean absolute PM error against the independent price at the true volatility is $0.318736 per barrel. At a cutoff of $0.005 per barrel, PI and PM differ by no more than the cutoff in 327 of 480 option states. Yet 315 of these 327 states have PM errors larger than the cutoff. This 96.3% figure depends on the chosen cutoff and simulated sample, which reuses histories across strikes. It is not an estimate of market risk. The result shows that PI and PM can agree closely even when both prices are inaccurate. It is separate from the test of the Taylor approximation to the PI–PM difference.

Table 1 compares three assumed correlations between futures contracts. The PI–PM difference is unchanged because those prices always assume $\rho = 1$. As the assumed correlation in the reference price decreases, plug-in errors increase and the nominal 95% posterior intervals contain fewer reference prices. These counts are unchanged when we increase the posterior grid from 513 to 1,025, 2,049, and 4,097 points. The comparison concerns the assumed model, not coverage of market prices or observed WTI correlations.

4.3 | Posterior integration in the WTI sample

The October-2026 sample contains 310 contract-date observations across 12 valuation dates. Table 2 shows nearly identical pricing errors for PI and PM (RMSE 0.5177 and 0.5183, respectively). Their mean absolute price difference is only 0.000745 dollars per barrel.

The central 95% posterior interval for conditional model prices has mean width 0.2733 dollars per barrel, approximately 367 times the mean absolute PI–PM difference. This shows why a small integration correction does not imply a narrow posterior distribution of model prices; it does not establish the economic value of that uncertainty. Both methods underprice October settlements on average.

4.4 | Historical versus option-implied volatility

Table 3 reports conditional pricing errors for 2,381 contract-date observations across 15 valuation dates. The historical PI benchmark has MAE 2.6187 and RMSE 3.4090, compared with 0.1075 and 0.1594 for the previous-day APO implied-volatility smile. The expanding earlier-date smile performs similarly. Both specifications reduce pricing errors for all seven expiries, with valuation-date cluster bootstrap intervals for the error differences below zero in each expiry (Figure 2).

Rolling-window, EWMA, and horizon-matched GARCH(1,1) benchmarks do not close the pricing-error gap (Online Supplement Section S1 and Table S1). In particular, GARCH gives MAE 2.6730 and RMSE 3.4441 on the same observations.

Panel B isolates 2,366 observations with an earlier implied volatility for the same contract. Carrying that

When posterior integration matters

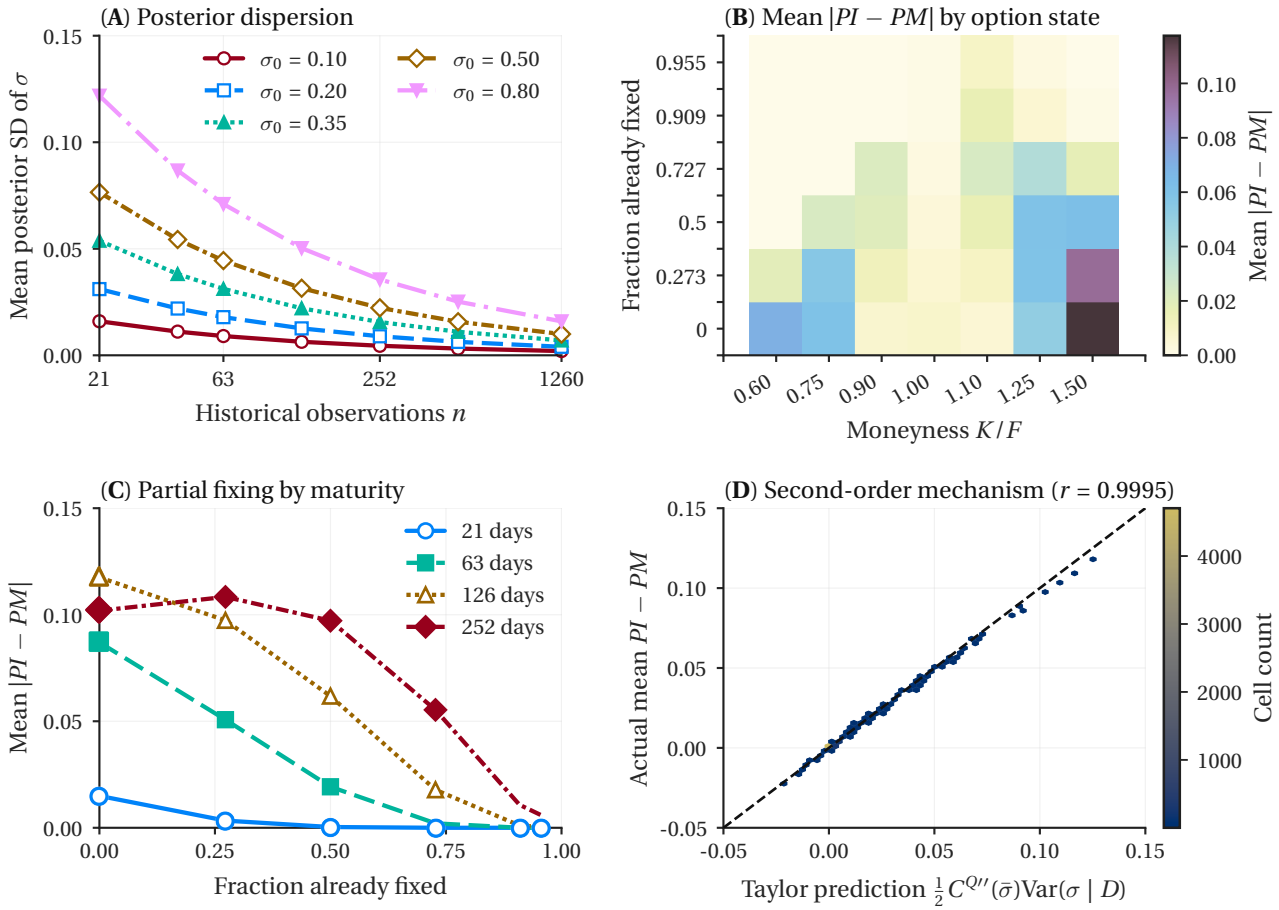


FIGURE 1 Posterior integration across 10 million simulated return histories and option states (random seed 2468). Panel A shows posterior dispersion by historical sample size. Panels B and C show the absolute PI–PM difference across moneyness and the realized-fixing fraction. Panel D compares mean PI–PM differences for 5,880 scenario cells with the second-order approximation in Equation (7); the cell-level correlation is 0.9995. The color bar shows the number of scenario cells per hexagonal bin.

TABLE 1 Simulation-based comparison of pricing error and posterior interval coverage. Values are in U.S. dollars per barrel. There are 160 independent return histories and 480 option-state observations per assumed correlation; strikes share return histories. Coverage intervals are 95% stratified bootstrap intervals resampling histories rather than option-state rows. The number of covered reference prices is unchanged across four posterior grids, although individual interval endpoints vary.

Reference ρ	Mean $\lvert PI - PM \rvert$	PM absolute error	95% coverage [bootstrap CI]
1.00	0.003709	0.318736	95.0% [91.3%, 98.1%]
0.75	0.003709	0.451675	65.2% [58.3%, 71.9%]
0.50	0.003709	0.721005	31.9% [26.9%, 36.9%]

volatility forward yields lower pooled errors than the fitted previous-day smile (MAE 0.0992 versus 0.1074). The lower errors relative to historical volatility do not depend on fitting a smile. The small difference between

TABLE 2 October-2026 WTI APO pricing and posterior price uncertainty. Pricing error is model price minus settlement price. All price quantities and errors are in U.S. dollars per barrel.

Quantity	N	Value
PI RMSE	310	0.5177
PM plug-in RMSE	310	0.5183
Mean $\lvert PI - PM \rvert$	310	0.000745
Mean 95% posterior price-interval width	310	0.2733

the two implied-volatility methods does not establish that one consistently outperforms the other.

4.5 | Cross-instrument comparison using WTI futures options

The cross-instrument comparison replaces APO-implied volatility with estimates from standard monthly WTI futures options. Table 4 uses 253 October-2026 APO contract-date observations across 10 valuation dates. Historical-volatility posterior integration has MAE 0.4464 and RMSE 0.5562. A previous-day vanilla-WTI implied-volatility specification based on LO options lowers these errors to 0.1559 and 0.2599. The corresponding previous-day APO implied-volatility smile has MAE 0.1676 and RMSE 0.2686.

Valuation-date cluster bootstrap results support the improvement over historical volatility. Differences between LO-implied and APO-implied specifications are small and do not establish consistent superiority by MAE and RMSE. The cross-instrument test instead shows that the advantage persists when implied volatility comes from another WTI option family.

5 | ROBUSTNESS AND INTERPRETATION

The main results change little under alternative assumptions about historical returns. A standardized Student-t model leaves the PI–PM difference small relative to historical-volatility pricing errors. Reconstructing first-nearby CL returns contract by contract gives a similar result: on the 2,381 date-ordered observations, historical PI has MAE 2.6468 and RMSE 3.4451, compared with 2.6187 and 3.4090 using the continuous/front-month return proxy. Further historical-volatility benchmarks appear in Online Supplement Section S1.

Numerical error does not explain the small empirical PI–PM difference. High-precision Monte Carlo, independently scrambled Sobol simulation, and Curran conditioning agree on the difference to a precision well below its observed magnitude. Simulation-based calibration provides a separate check on the posterior implementation. The supplement compares a plug-in price using the posterior root-mean-square volatility (Section S2), alternative ways to convert LO-implied volatilities to one APO input, and the accuracy of the Cox–Ross–Rubinstein tree (Section S4). These numerical checks do not establish an exact arbitrage-free representation of the volatility surface.

The independent numerical check confirms that Curran's approximation does not give the exact arithmetic-average option price. In six one-factor numerical controls, its largest absolute difference from the scrambled Sobol reference is \$0.000922 per barrel. The largest corresponding half-width of the 95% reference confidence interval is \$0.000154 per barrel. When $\rho = 0.50$, the largest reference confidence-interval half-width is approximately \$0.001023 per barrel, much smaller than the changes in the simulated reference prices. Repeating the calculation with 513, 1,025, 2,049, and 4,097 volatility-grid points did not change which of the 1,440 synthetic observations were covered. The covered counts were 456, 313, and 153 out of 480 for $\rho = 1$, 0.75, and 0.50, respectively. Against the 4,097-node benchmark, the largest 513-node PI difference is below \$0.000000005 per barrel and the largest PM difference is \$0.000056 per barrel, but posterior interval endpoints differ by up to \$0.005227 per barrel. For one case with $\rho = 0.75$, the reference price's QMC uncertainty interval overlaps a posterior interval boundary on the two finest grids. The coverage result based on the point reference does not change. Coverage counts are stable across these grids. This does not prove convergence to the exact continuous-grid result or calibration to market prices. In particular, results at a \$0.001 decision threshold do not establish that payoff prices are accurate to \$0.001. We designed the follow-up after examining the pilot, and we did not preregister it. The assumed correlations are hypothetical; they do not measure misspecification in WTI futures or validate exchange prices.

The market comparison also has limitations. In the original 2,381-observation comparison, 2,319 option-date observations have zero recorded trading volume. Implied volatility still gives lower errors when we restrict the sample by trading volume or open interest,

TABLE 3 Historical-volatility and prior option-implied volatility benchmarks. Panel A uses 2,381 date-ordered contract-date observations on 15 valuation dates. Panel B restricts the comparison to observations for which the same contract has a strictly prior implied volatility. Historical PI is priced by Monte Carlo; the other historical, GARCH, and implied-volatility estimates are priced with the same Curran approximation. MAE and RMSE are in U.S. dollars per barrel.

Panel A: date-ordered conditional pricing comparison

Volatility input	*N*	Dates	MAE	RMSE
Historical PI	2,381	15	2.6187	3.4090
Rolling 63 returns	2,381	15	4.5201	5.6465
Horizon-matched GARCH(1,1)	2,381	15	2.6730	3.4441
Previous-day APO implied-volatility smile	2,381	15	0.1075	0.1594

Panel B: same-contract lagged-IV benchmark

Volatility input	*N*	Dates	MAE	RMSE
Historical PI	2,366	15	2.6279	3.4173
Same-contract lagged IV	2,366	15	0.0992	0.1403
Previous-day fitted APO smile	2,366	15	0.1074	0.1594

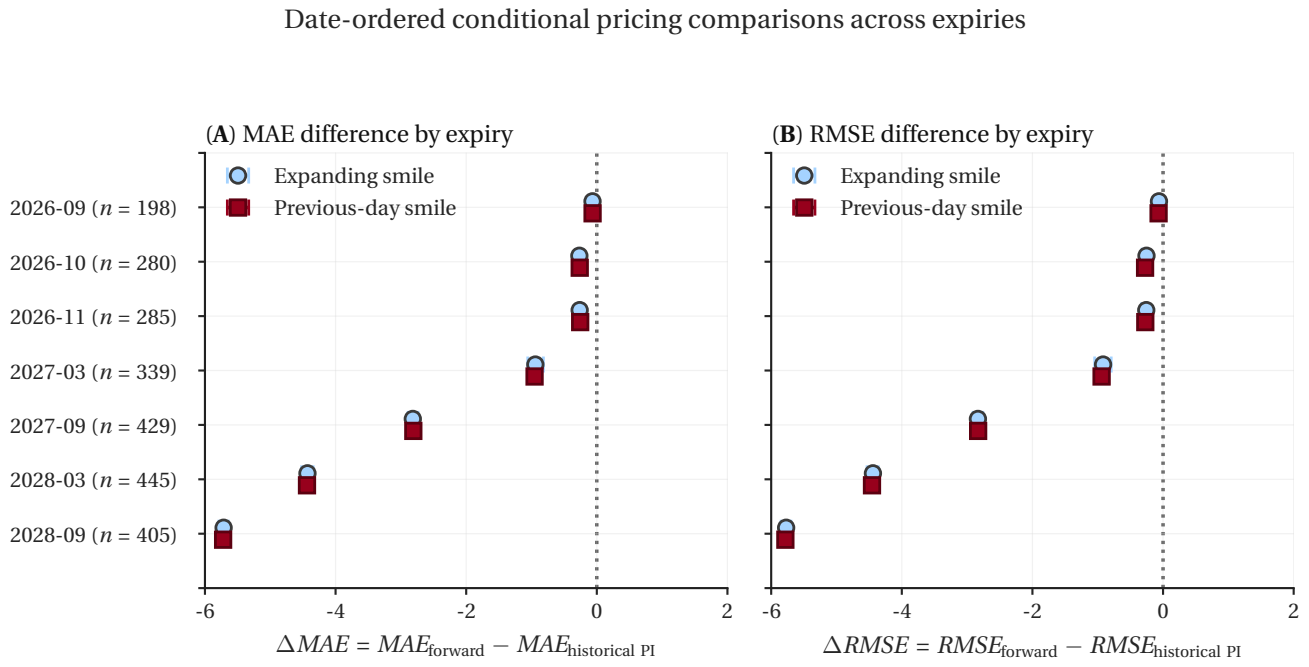


FIGURE 2 Date-ordered conditional pricing-error differences by APO expiry. Points compare the earlier-date APO implied-volatility specifications with the historical posterior-integrated benchmark; negative values indicate lower error for the implied-volatility specification. Horizontal bars are 95% percentile intervals from valuation-date cluster bootstrap resampling. All implied-volatility inputs are dated before the valuation date in the archived records; original publication times and subsequent revisions are not independently verified.

although some expiry groups are small (Online Supplement Section S3). Moreover, the main pricing comparison contains only 15 valuation dates and the external comparison only 10. We cannot independently verify when the vendor first published or later revised its records. The valuation-date cluster bootstrap accounts for dependence among contracts on the same date, but not fully for dependence between nearby dates.

An additional concern is that CME Clearing began using Curran's model for WTI Average Price Option

TABLE 4 Cross-instrument comparison of 253 October-2026 APO contract-date observations whose moneyness falls within the range observed in earlier LO prices. Historical PI uses Monte Carlo; the vanilla-WTI and APO implied-volatility inputs use Curran's approximation. All implied-volatility inputs use only earlier-date option information. MAE and RMSE are in U.S. dollars per barrel.

Volatility input	MAE	RMSE
Historical PI	0.4464	0.5562
Previous-day vanilla-WTI IV	0.1559	0.2599
Previous-day APO IV	0.1676	0.2686

end-of-day settlement valuation on May 18, 2020 (CME Clearing 2020). As a result, Curran-based model prices and settlement-price proxies may reflect some of the same valuation assumptions. The Barchart `Latest` field has not been independently tied to the exchange's final settlement calculation. The comparison therefore measures agreement with available end-of-day price proxies under the assumed pricing model. It does not independently validate the market's price dynamics.

Finally, the one-factor constant-volatility model excludes richer futures-curve dynamics, stochastic volatility, jumps, and convenience yields. Its implied volatilities are model-equivalent pricing inputs, not uniquely identified risk-neutral diffusion parameters or direct estimates of a variance risk premium. The pricing-error differences apply to these inputs and this model; they do not identify physical and risk-neutral volatility components, trading gains, or the value of a hedging strategy.

6 | CONCLUSION

Posterior integration and the choice of volatility estimate affect WTI APO prices in different ways. The simulations show that PI and PM can differ when volatility is uncertain and the pricing function has substantial curvature with respect to volatility. In the October-2026 WTI sample, however, the PI–PM difference is small even though the posterior distribution of model prices remains much wider. Close agreement between PI and PM therefore does not mean that model-price uncertainty is small.

A separate simulation shows that PI and PM can agree even when both are far from the price calculated at the true volatility. When we lower the assumed correlation between futures contracts in the independent pricing reference, fewer reference prices fall within the posterior intervals, even though PI and PM remain unchanged. The coverage rates are stable across four numerical grids, but they do not show how well the intervals cover observed market prices.

In the conditional APO pricing comparisons, earlier-date option-implied volatility gives lower errors against settlement-price proxies than the tested historical-volatility benchmarks. The result persists without cross-sectional smile fitting and when implied volatility is obtained from standard monthly WTI futures options. For the one-factor model and valuation dates studied, the choice of volatility estimate matters more for agreement with settlement-price proxies than posterior integration alone. These comparisons do not establish fully prospective performance or a general advantage over richer commodity models.

DATA AVAILABILITY

We obtained the WTI APO and CL histories through a commercial Barchart subscription. End-of-day `Latest` prices are treated as settlement-price proxies, not independently verified transaction prices. The separate comparison with standard WTI futures options uses CME/NYMEX LO and CL data from Databento. These market observations are subject to third-party licensing and data-use restrictions. Proprietary raw datasets and restricted row-level records are not published on Zenodo or GitHub; researchers must arrange authorized access directly with the data providers. Internal market-data provenance and restricted empirical analyses remain in the private research repository.

CODE AVAILABILITY

A versioned public archive of selected research software, tests, documentation, and synthetic benchmark results is preserved on Zenodo. Its DOI and repository URL are supplied on the separate identifying title page and omitted here to preserve double-anonymized review. The public archive does not contain licensed Barchart or Databento observations, the complete private research repository, or an independently verified reproduction of all commercial-data empirical results. Additional internal source and analysis records remain private and may be made available confidentially to the editor, subject to applicable restrictions.

PLAIN LANGUAGE SUMMARY

For WTI crude-oil average-price options, allowing for uncertainty about estimated volatility may change a model price only slightly. That does not mean the price

estimate is accurate or that volatility uncertainty is small. We compare two ways to use the same volatility posterior and examine how much the choice of historical or earlier option-implied volatility affects conditional pricing errors.

AI-ASSISTED PREPARATION DISCLOSURE

OpenAI ChatGPT (models GPT-5.4, GPT-5.5, GPT-5.6, and GPT-6) was used during manuscript preparation and revision in 2026, including SeptemberNovember 2026. It assisted with research software development and debugging, literature and experimental organization, and drafting and language editing of the abstract, introduction, methods, results discussion, and online supplement. The author reviewed AI-assisted code using repository tests and reproducible computational checks, and checked textual suggestions, citations, and numerical statements against source literature and saved results. The author made all scientific decisions and remains responsible for the methods, analyses, claims, and final manuscript. The study used commercially licensed Barchart and Databento data. No proprietary Barchart or Databento market data were entered into any AI service. Their proprietary raw records are not redistributed through the manuscript or public research outputs, and no human-subject data were used. Compliance with vendor licenses and AI-service data-handling requirements remains the author's responsibility.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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